

In [1]: `import pandas as pd`

In [3]: `df = pd.read_csv("Titanic-Dataset.csv")`

In [4]: `df.head()`

Out[4]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>PassengerId</th> <th>Survived</th> <th>Pclass</th>
<th>Name</th> <th>Sex</th> <th>Age</th> <th>SibSp</th> <th>Parch</th>
<th>Ticket</th> <th>Fare</th> <th>Cabin</th> <th>Embarked</th> </thead>
<tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody>
<tbody></tbody>

1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [5]: `df.describe()`

Out[5]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead>
<th></th> <th>PassengerId</th> <th>Survived</th> <th>Pclass</th> <th>Age</th>
<th>SibSp</th> <th>Parch</th> <th>Fare</th> </thead> <tbody> <th>count</th>
<th>mean</th> <th>std</th> <th>min</th> <th>25%</th> <th>50%</th> <th>75%</th>
<th>max</th> </tbody> <tbody></tbody>

891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [6]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
In [7]: df["Survived"].value_counts()
```

```
Out[7]: Survived
0      549
1      342
Name: count, dtype: int64
```

```
In [8]: df.isnull().sum()
```

```
Out[8]: PassengerId    0
Survived             0
Pclass              0
Name                0
Sex                 0
Age                177
SibSp              0
Parch              0
Ticket             0
Fare               0
Cabin             687
Embarked           2
dtype: int64
```

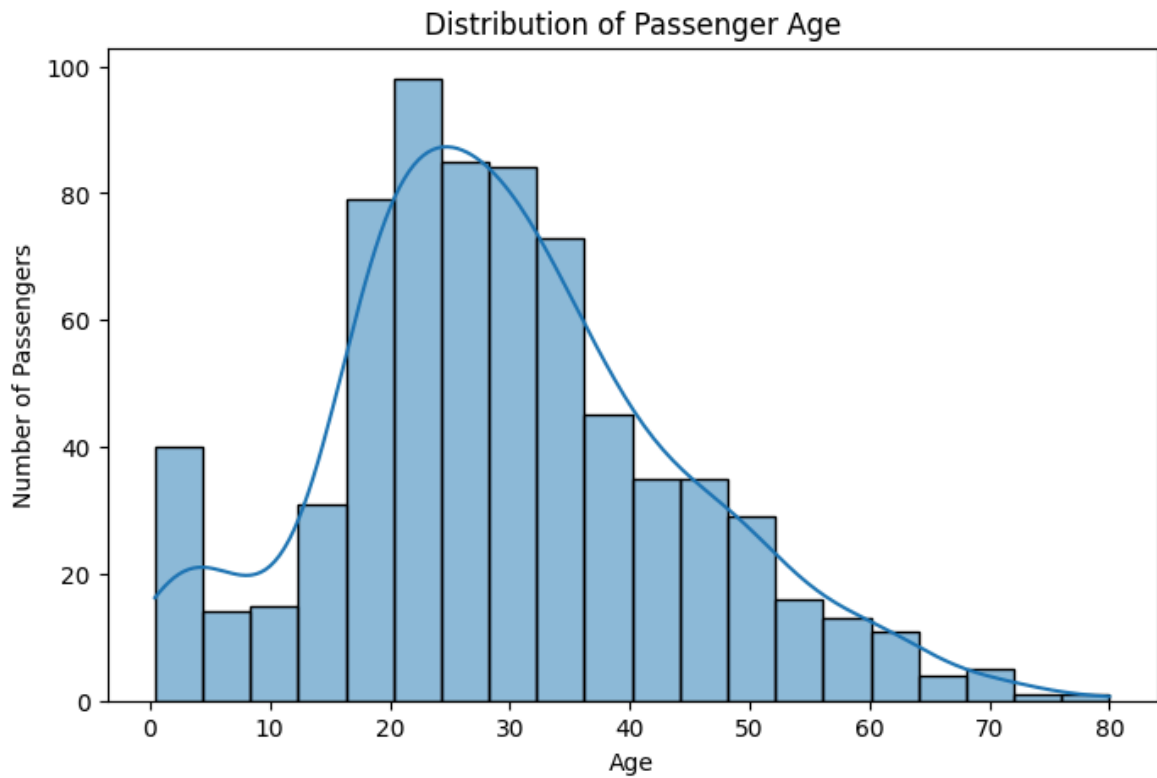
```
In [9]: plt.figure(figsize=(8,5))
sns.histplot(df["Age"].dropna(), bins=20, kde=True)
plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.show()
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[9], line 1
----> 1 plt.figure(figsize=(8,5))
      2 sns.histplot(df["Age"].dropna(), bins=20, kde=True)
      3 plt.title("Distribution of Passenger Age")
      4 plt.xlabel("Age")

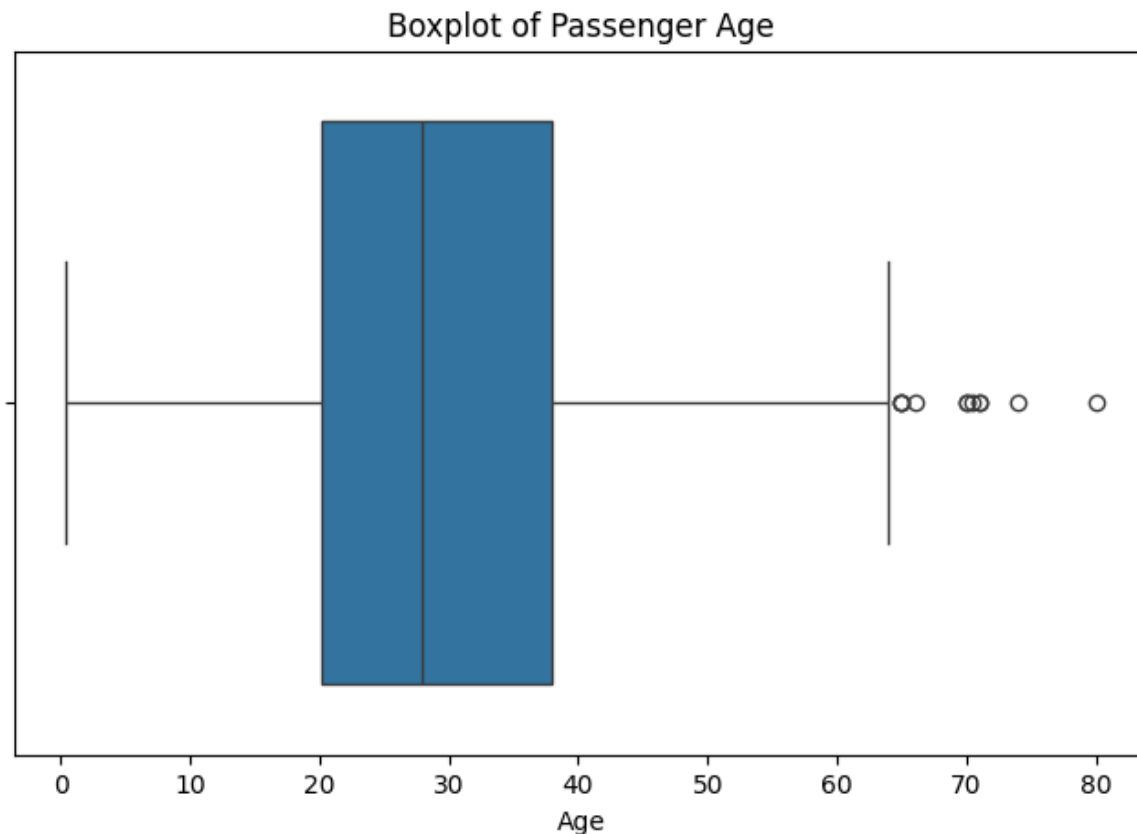
NameError: name 'plt' is not defined
```

```
In [10]: import matplotlib.pyplot as plt  
import seaborn as sns
```

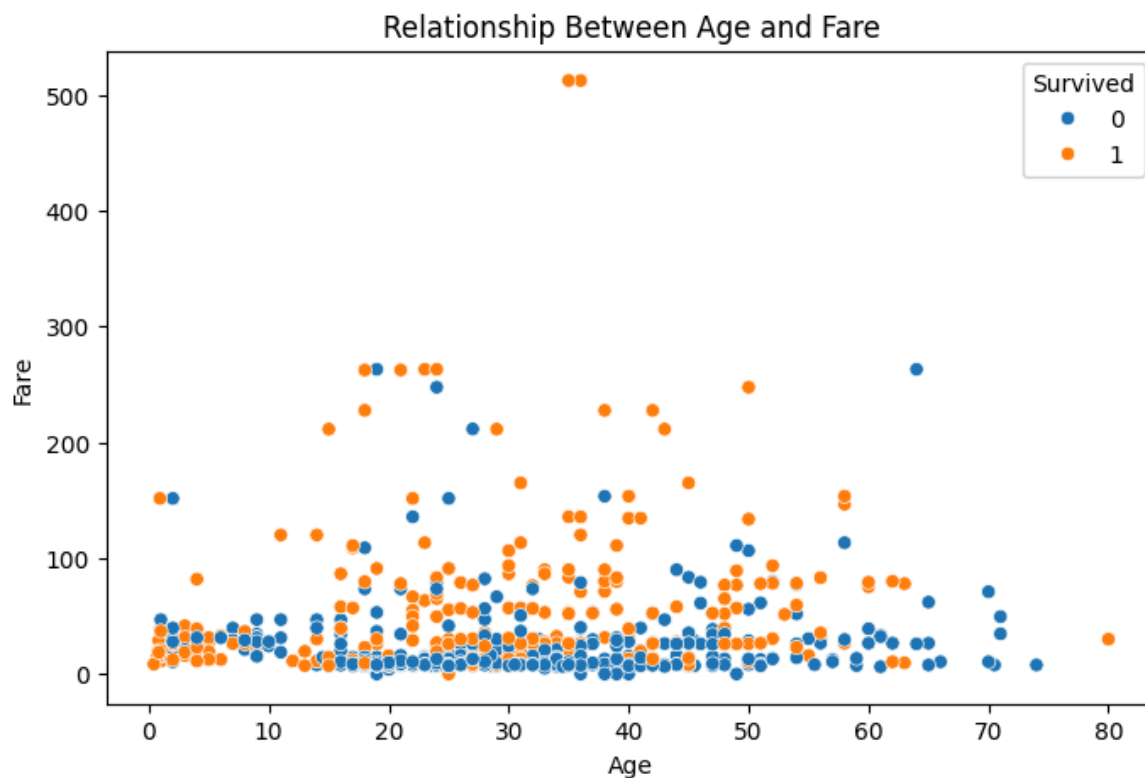
```
In [11]: plt.figure(figsize=(8,5))  
sns.histplot(df["Age"].dropna(), bins=20, kde=True)  
plt.title("Distribution of Passenger Age")  
plt.xlabel("Age")  
plt.ylabel("Number of Passengers")  
plt.show()
```



```
In [12]: plt.figure(figsize=(8,5))  
sns.boxplot(x=df["Age"].dropna())  
plt.title("Boxplot of Passenger Age")  
plt.xlabel("Age")  
plt.show()
```

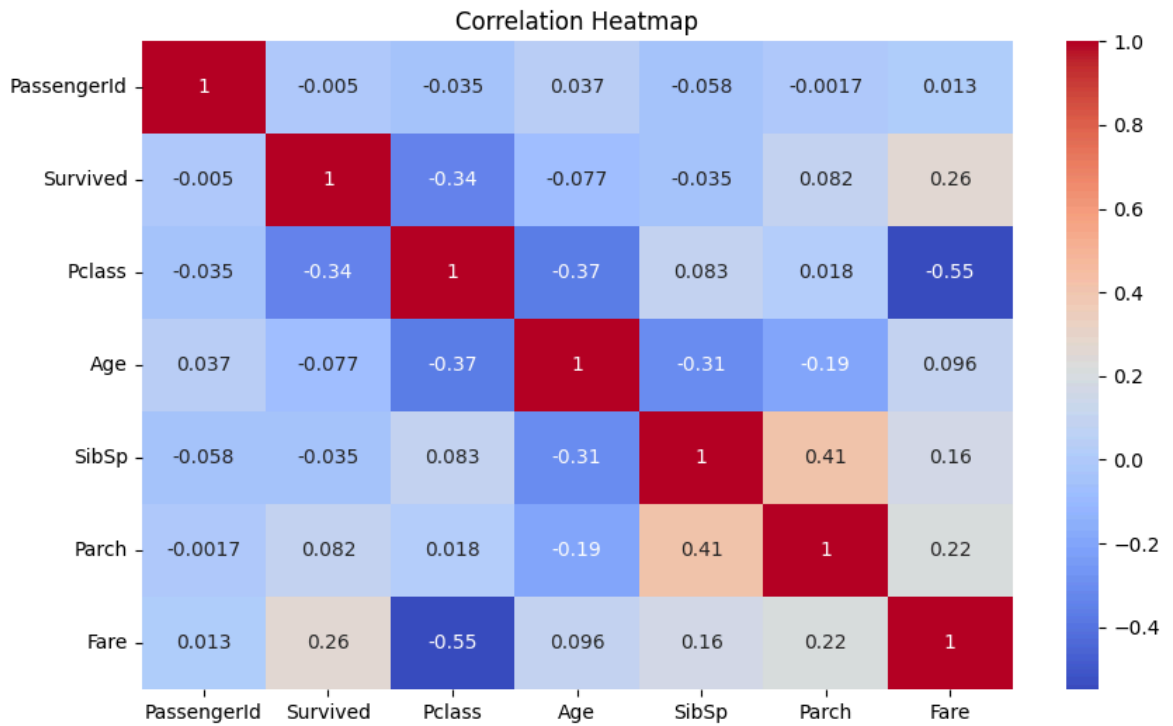


```
In [13]: plt.figure(figsize=(8,5))
sns.scatterplot(data=df, x="Age", y="Fare", hue="Survived")
plt.title("Relationship Between Age and Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

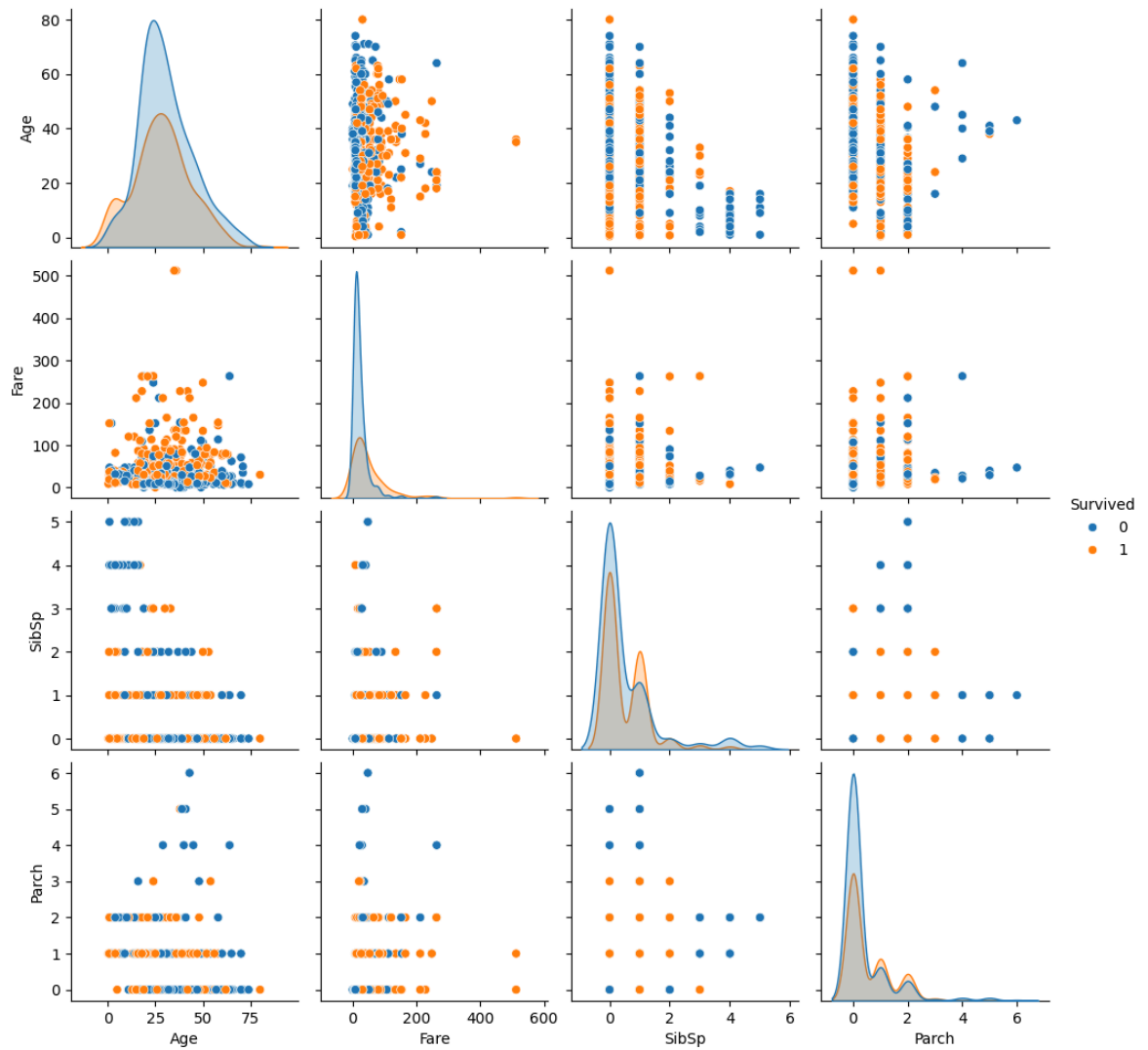


```
In [14]: plt.figure(figsize=(10,6))
sns.heatmap(df.select_dtypes(include="number").corr(), annot=True, cmap="coolwar
```

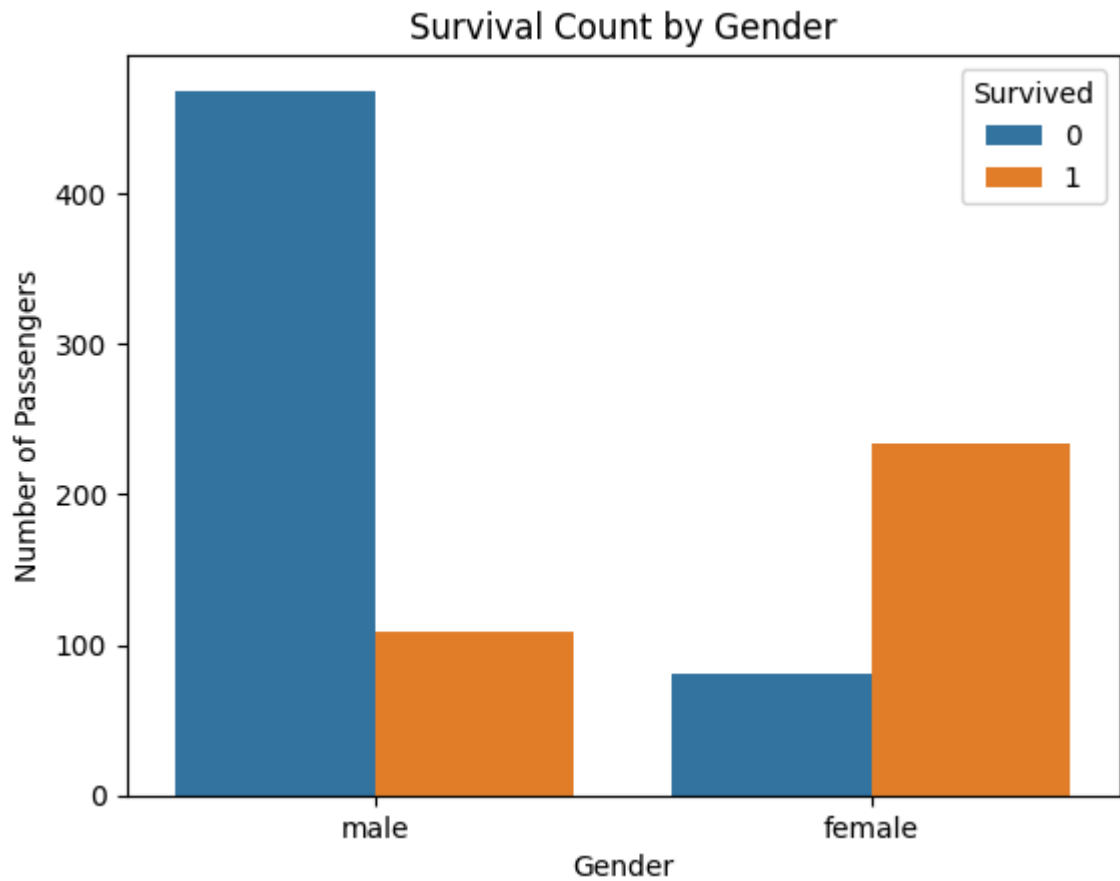
```
plt.title("Correlation Heatmap")
plt.show()
```



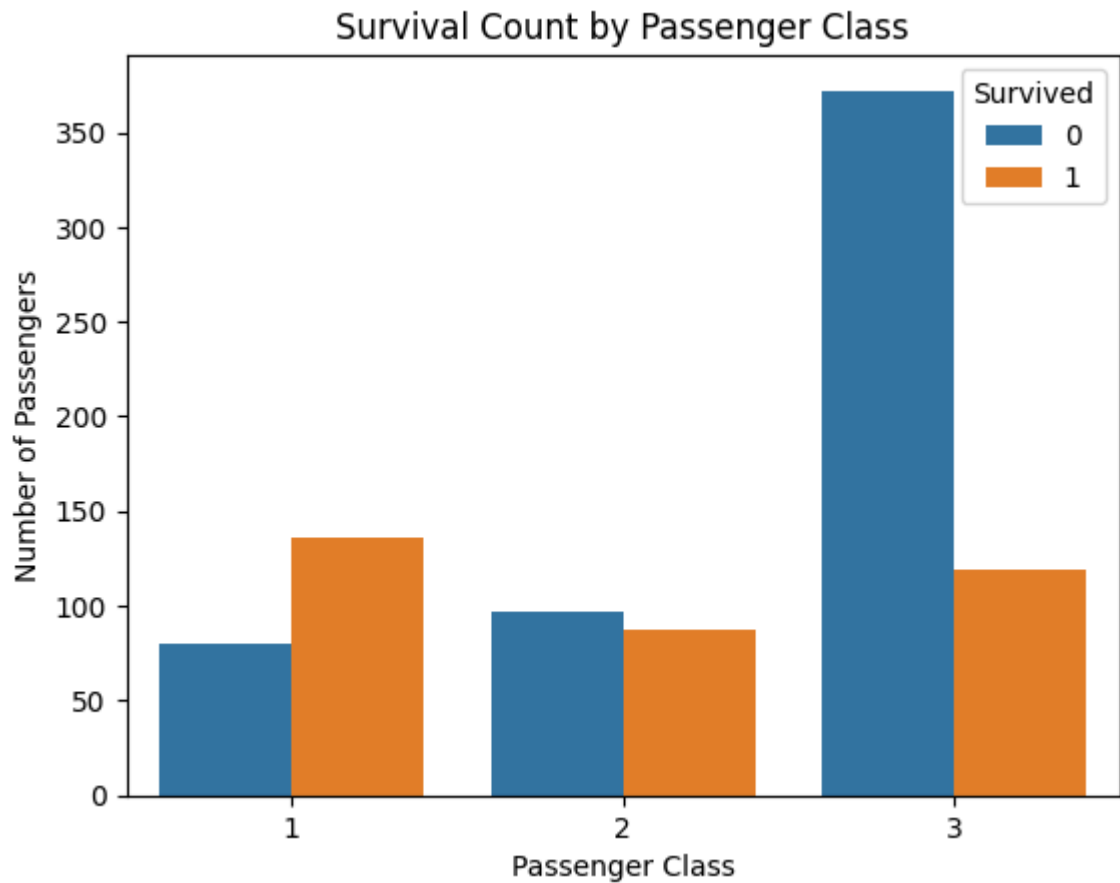
```
In [15]: sns.pairplot(
            df[["Survived", "Age", "Fare", "SibSp", "Parch"]].dropna(),
            hue="Survived"
        )
plt.show()
```



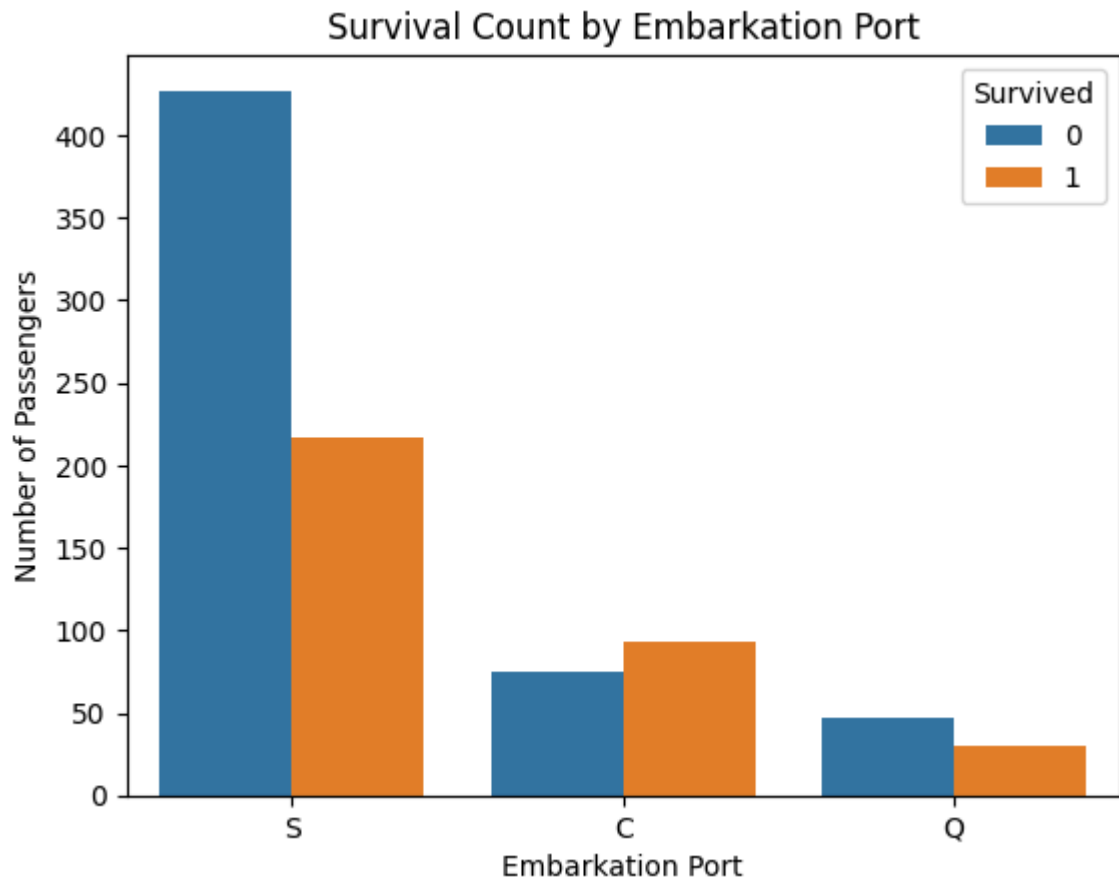
```
In [16]: sns.countplot(data=df, x="Sex", hue="Survived")
plt.title("Survival Count by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.show()
```



```
In [17]: sns.countplot(data=df, x="Pclass", hue="Survived")
plt.title("Survival Count by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()
```



```
In [18]: sns.countplot(data=df, x="Embarked", hue="Survived")
plt.title("Survival Count by Embarkation Port")
plt.xlabel("Embarkation Port")
plt.ylabel("Number of Passengers")
plt.show()
```

```
In [19]: df.groupby("Sex")["Survived"].mean() * 100
```

```
Out[19]: Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64
```

```
In [20]: print("Observation:")
print("The age distribution shows that passengers were mostly young and middle-a
print("There are fewer passengers at very old ages.")
print("The distribution also contains some missing age values, which were exclud
```

Observation:

The age distribution shows that passengers were mostly young and middle-aged.

There are fewer passengers at very old ages.

The distribution also contains some missing age values, which were excluded from the histogram.

```
In [21]: print("Observation:")
print("The boxplot shows the central range and spread of passenger ages.")
print("Some passengers are at unusually high ages, which may appear as outliers.
print("The median age is around the middle of the overall age distribution.")
```

Observation:

The boxplot shows the central range and spread of passenger ages.

Some passengers are at unusually high ages, which may appear as outliers.

The median age is around the middle of the overall age distribution.

```
In [22]: print("Observation:")
print("The scatterplot shows the relationship between passenger age and fare.")
print("There is no strong direct relationship between age and fare.")
print("Passengers who paid very high fares are visible as points at the upper pa
print("Survival status shows different patterns across age and fare values.")
```

Observation:

The scatterplot shows the relationship between passenger age and fare.

There is no strong direct relationship between age and fare.

Passengers who paid very high fares are visible as points at the upper part of the plot.

Survival status shows different patterns across age and fare values.

```
In [23]: print("Observation:")
print("The heatmap shows the correlation between numerical variables in the data")
print("Survived has a positive relationship with Fare and Pclass is negatively related to Survived.")
print("SibSp and Parch show a positive relationship with each other.")
print("Most other numerical variables have relatively weak correlations.")
```

Observation:

The heatmap shows the correlation between numerical variables in the dataset.

Survived has a positive relationship with Fare and Pclass is negatively related to Survived.

SibSp and Parch show a positive relationship with each other.

Most other numerical variables have relatively weak correlations.

```
In [24]: print("Observation:")
print("The pairplot shows relationships among Age, Fare, SibSp, Parch, and Survived.")
print("Fare shows some separation between survived and non-survived passengers.")
print("Age does not show a strong clear separation based on survival.")
print("Most numerical variables show weak to moderate relationships.")
```

Observation:

The pairplot shows relationships among Age, Fare, SibSp, Parch, and Survived.

Fare shows some separation between survived and non-survived passengers.

Age does not show a strong clear separation based on survival.

Most numerical variables show weak to moderate relationships.

```
In [25]: print("Observation:")
print("The graph shows a clear difference in survival between male and female passengers.")
print("Female passengers had a higher survival rate than male passengers.")
print("Male passengers had a higher number of non-survivors.")
```

Observation:

The graph shows a clear difference in survival between male and female passengers.

Female passengers had a higher survival rate than male passengers.

Male passengers had a higher number of non-survivors.

```
In [26]: print("Observation:")
print("The graph shows a clear difference in survival across passenger classes.")
print("First-class passengers had a higher number of survivors compared with second and third class.")
print("Third-class passengers had the highest number of non-survivors.")
```

Observation:

The graph shows a clear difference in survival across passenger classes.

First-class passengers had a higher number of survivors compared with second and third class.

Third-class passengers had the highest number of non-survivors.

```
In [27]: print("Observation:")
print("The graph compares survival outcomes across the three embarkation ports.")
print("Passengers who embarked from Southampton form the largest group.")
print("The number of survivors and non-survivors varies across embarkation ports")
```

Observation:

The graph compares survival outcomes across the three embarkation ports.

Passengers who embarked from Southampton form the largest group.

The number of survivors and non-survivors varies across embarkation ports.

```
In [28]: print("Overall EDA Summary:")
print("1. The Titanic dataset contains numerical and categorical passenger infor
print("2. Age has a wide distribution, with most passengers being young and midd
print("3. Female passengers had a higher survival rate than male passengers.")
print("4. Passenger class had a strong influence on survival, with first-class p
print("5. Fare shows some relationship with survival, with higher fares generall
print("6. The correlation heatmap shows relationships among the numerical variab
print("7. The analysis shows that gender, passenger class, and fare were importa
```

Overall EDA Summary:

1. The Titanic dataset contains numerical and categorical passenger information.
2. Age has a wide distribution, with most passengers being young and middle-aged.
3. Female passengers had a higher survival rate than male passengers.
4. Passenger class had a strong influence on survival, with first-class passengers having better survival outcomes.
5. Fare shows some relationship with survival, with higher fares generally associated with better survival outcomes.
6. The correlation heatmap shows relationships among the numerical variables.
7. The analysis shows that gender, passenger class, and fare were important factors associated with survival.

```
In [ ]:
```